

Jenkins: The Heart of DevOps

Jenkins is a powerful open-source Continuous Integration/Continuous Delivery (CI/CD) tool, widely used in DevOps workflows. It is plugin-based, which means you can add various plugins to extend its functionality, making it highly flexible and customizable.

Key Concepts in Jenkins:

- **Jobs:** A job in Jenkins is a set of instructions that Jenkins executes. These can be build jobs, test jobs, or deployment tasks.
- **Plugins:** Plugins in Jenkins add extra functionalities, such as integrating with Git, Docker, or other CI/CD tools.

Steps to Install Jenkins on AWS (Red Hat-based Linux Distribution):

1. **Create a Server:** Launch an EC2 instance on AWS (using a Linux AMI like Amazon Linux 2 or CentOS).
2. **Access Your Server:**
 - Connect to your EC2 instance using SSH.
 - Use the `ssh` command with your PEM file to access the server.
Example: `ssh -i "your-key.pem" ec2-user@<your-ec2-public-ip>`
3. **Switch to Root User:** Once logged in, switch to the root user to execute installation commands:
Example: `sudo su`
4. **Install Jenkins:** Follow these commands to install Jenkins on your server. You can refer to the [official Jenkins documentation](#) for the latest steps.

```
# Update your system packages
sudo yum update -y
```

```
# Add Jenkins repository
sudo wget -O /etc/yum.repos.d/jenkins.repo https://pkg.jenkins.io/redhat-stable/jenkins.repo
```

```
# Import the Jenkins key
sudo rpm --import https://pkg.jenkins.io/redhat-stable/jenkins.io-2023.key
```

```
# Install Java (required for Jenkins)
sudo dnf install java-17-amazon-corretto -y
```

```
# Install Jenkins
sudo yum install jenkins -y
```

```
# Enable Jenkins to start on boot
sudo systemctl enable jenkins
```

```
# Start Jenkins service
sudo systemctl start jenkins
```

```
# Check Jenkins status to ensure it's running
sudo systemctl status jenkins
```

5. Access Jenkins:

- After successful installation, you can access Jenkins through your server's public IP using port **8080**. Open a web browser and enter:

Example: `http://<your-ec2-public-ip>:8080`

6. Unlock Jenkins:

- Jenkins will prompt you for an admin password to unlock it. You can retrieve this password from the following file:

Example: `sudo cat /var/lib/jenkins/secrets/initialAdminPassword`

7. Install Plugins and Set Up Jenkins:

- After unlocking Jenkins, it will prompt you to install recommended plugins. Follow the on-screen instructions to complete the installation.
- Create a new admin user by providing a username and password.

8. Jenkins Dashboard:

- Once the setup is complete, you will land on the Jenkins dashboard. From here, you can start creating jobs, managing plugins, and configuring your CI/CD pipelines.

Additional Tips:

- **Automation:** You can automate these installation steps by creating a shell script with all the above commands and running it in one go.
- **Backup:** Regularly back up your Jenkins configuration and job data, especially if you are running critical CI/CD pipelines.
- **Security:** Ensure Jenkins is secured with appropriate authentication mechanisms, and consider enabling SSL for secure access.

By following these steps, you can easily set up Jenkins on AWS and begin leveraging its power for automating your software delivery processes.